Ziyu Wang

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Research Interests: Vision-and-Language Navigation, Robot Learning, Knowledge Distillation, Vision-Language-Action Models

Education

Southern University of Science and Technology (SUSTech) 2024 - Present
Undergraduate, Computer Science (Turing Class)

GPA: 3.73/4.00

Coursework: Computer Vision, Artificial Intelligence (H), Algorithms, Operating Systems, Computer Networks.

Research Experience

Panorama-OPSD: On-Policy Self-Distillation for VLN 2026 - Present

- Designing a two-stage teacher-student framework for continuous navigation, using panoramic context during training and only monocular RGB, visual history, and language instructions at deployment.
- The design first jointly fine-tunes a shared policy on paired panoramic and monocular observations with identical expert-action labels, then freezes a copy in panoramic mode as the teacher.
- On-policy self-distillation trains the monocular student on its own sampled action sequences, using the panoramic teacher's next-token distributions under the same generated prefixes to guide the student's own action choices.

Failure Analysis and Test-Time Adaptation for VLN 2026 - Present

- Conducted in-depth literature reviews and failure-case analyses of several VLN foundation models.
- Exploring self-supervised test-time parameter updates using only monocular RGB observations, visual history, and language instructions for reliable navigation adaptation.
- Trained and evaluated scorers for eight action candidates from frozen NaVIDA, using instruction, visual-history, and ordered action features with candidate-conditioned spatial attention.
- Compared success-prediction and pairwise-ranking objectives, action-only and state-action baselines, and mismatched-input controls on R2R-CE training scenes.

Robotics Project

RescueScout: Confidence-Aware Search-and-Report Robot 2026
National University of Singapore

- Contributed to integrating a Raspberry Pi-Arduino mobile robot with camera-based perception, teleoperation, ultrasonic safety sensing, and shared autonomy.
- Contributed to marker-assisted topological mapping and route planning for semi-autonomous navigation, including confidence-aware perception and human review of uncertain detections.

Technical Skills

Programming	Python, C++, Java
Frameworks and Libraries	PyTorch, NumPy, OpenCV, Matplotlib
Simulation	Habitat

Honors and Competitions

- **Honors Student**, Turing Class, SUSTech.
- **Outstanding Scholarship**, SUSTech: Second Prize (sophomore year); Third Prize.
- **Award of Excellence**, Guangdong Undergraduate Mathematical Contest in Modeling, 2025.
- Successful Participant, Mathematical Contest in Modeling (MCM), COMAP, 2026.